

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Third Semester B.Tech (CBCS) Degree Examinations

December 2025

EE2005-1 – NETWORK ANALYSIS
 (For EE)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five** full questions choosing **Two** full questions from **Unit – I & Unit – II** each and **One** full question from **Unit – III**.

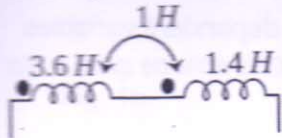
PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. KCL works on the principle of which of the following

- A) law of conservation of charge
 B) law of conservation of energy.
 C) Both
 D) None of these

2. The equivalent inductance is

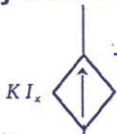
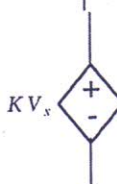
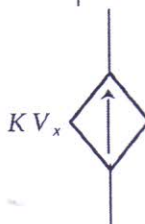


- A) 4 H
 B) 6 H
 C) 7 H
 D) 0 H

3. Connecting two voltage sources of unequal ratings in series results in —

- A) Average of voltages
 B) Algebraic sum of voltages
 C) Zero resultant voltage
 D) Circulating current and damage

4. The symbol of CCVS

- A) 
 B) 
 C) 
 D) 

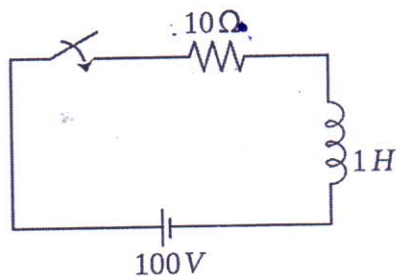
5. In a three phase, delta connection —

- A) line current is equal to phase current
 B) Line voltage is equal to phase voltage
 C) Line voltage and line current is zero
 D) Line voltage is equal to square root 3 times phase voltage

6. Reactive power is expressed in?

- A) Watts
 B) VAR
 C) VA
 D) No unit

7. In a series RLC circuit that is operating above the resonant frequency, the current
 A) Is in phase with the applied voltage B) Lags the applied voltage
 C) Leads the applied voltage D) None of these
8. The total reactance of series RLC circuit at resonance is
 A) Equal to X_L B) Equal to X_C
 C) Equal to R D) Zero
9. The s-domain equivalent of the capacitor is ?
 A) C B) C/s
 C) $1/Cs$ D) s/C
10. The unenergized capacitor at $t=0^-$ behaves as -----at $t=0^+$
 A) short circuit B) open circuit
 C) voltage source D) current source
11. Inductor does not allow sudden change in
 A) Voltage B) Frequency
 C) Phase D) Current
12. If $f(t)=r(t-a)$, $F(s)$ is-----
 A) $\frac{e^{-as}}{s^2}$ B) $\frac{e^{-as}}{s}$
 C) $\frac{e^{as}}{s^2}$ D) $\frac{e^{as}}{s}$
13. The h-parameter model is called a hybrid model because
 A) It uses both impedance and admittance terms B) It uses both voltage and current as dependent variable
 C) It applies only to hybrid circuits D) It uses mixed units for parameters
14. The admittance (Y) parameters of a two-port network are defined under
 A) Short-circuit output conditions B) Open-circuit output conditions
 C) Short-circuit input conditions D) Short-circuit output and input conditions
15. If the switch is closed at $t=0$, what is the current in the circuit at $t=0^+$?



- A) 0 A B) 10 A
 C) 20 A D) Infinity
16. The highest power of s in the equation will be equal to
 A) Number of sources in the circuit B) Total number of elements in the circuit
 C) Number of heat dissipation element in the circuit D) Number of independent energy storage elements
17. Thevenin's equivalent circuit consists of a _____
 A) Voltage source in series with a resistor B) Current source in parallel with a resistor
 C) Voltage source in parallel with a resistor D) Current source in series with a resistor

18. "In any linear bilateral network, if a source of emf V volt in any branch produces a current I in any other branch, then same emf acting in the second branch would produce same current in the first branch". The above statement is associated with
- A) Thevenin's Theorem B) Superposition Theorem
C) Reciprocity Theorem D) Norton's Theorem
19. If source impedance is complex, then maximum power transfer occurs when the load impedance is _____ the source impedance.
- A) Equal to B) Negative of
C) Complex conjugate of D) Negative of complex conjugate of
20. Millman's theorem yields
- A) Equivalent resistance B) Equivalent impedance
C) Equivalent voltage source D) Equivalent voltage or current source

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

Marks BT* CO* PO*

1. a) Using super node technique find v_1 in the circuit in shown in Fig. 1a.

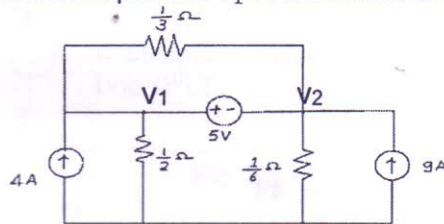


Fig. 1a.

4 L*3 1 1,2

- b) Using source transformation technique find V for the circuit shown in Fig. 1b.

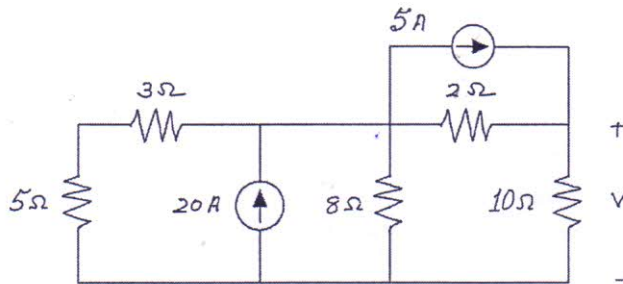


Fig. 1b.

4 L3 1 1,2

- c) An inductive coil of resistance 6Ω and inductance $1mH$ is connected in parallel with another branch consisting of a resistance of 4Ω in series with a capacitance of $20\mu F$. Find the resonant frequency and the corresponding current when the applied voltage is $200V$.

8 L3 2 1,2

2. a) Derive the expression for the neutral shift voltage in a three-phase unbalanced load (Neutral disconnected).
- b) Obtain the expression for natural frequency, quality factor and bandwidth for a series RLC circuit.
- c) Compute the currents i_1 and i_2 for the circuit shown in Fig. 2c.

4 L3 2 1,2

6 L2 2 1,2

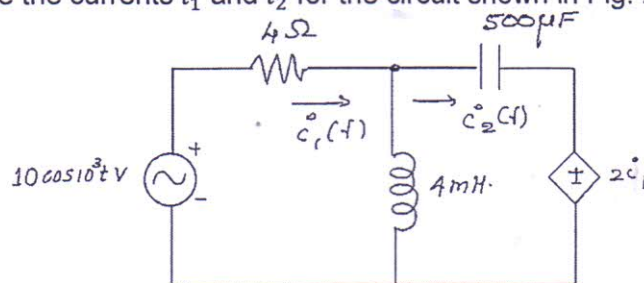


Fig. 2c.

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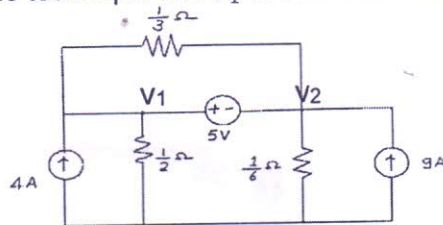


Fig. 1a.

4 L*3 1 1,2

- b) Using source transformation technique find V for the circuit shown in Fig. 1b.

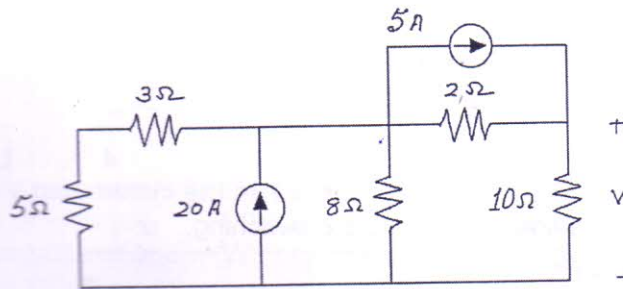


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- c) An inductive coil of resistance 6Ω and inductance $1mH$ is connected in parallel with another branch consisting of a resistance of 4Ω in series with a capacitance of $20\mu F$. Find the resonant frequency and the corresponding current when the applied voltage is $200V$.

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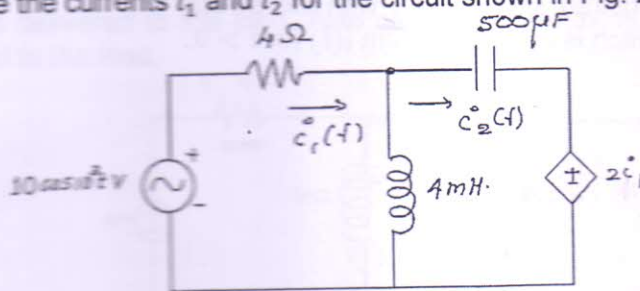


Fig. 2c.

6 L3 1 1,2

3. a) State the conditions when super-node or super-mesh technique to be used while solving the given network.
 b) Calculate the loop currents in the circuit shown in Fig. 3b. when $v(t) = 128 \sin(1000t) V$

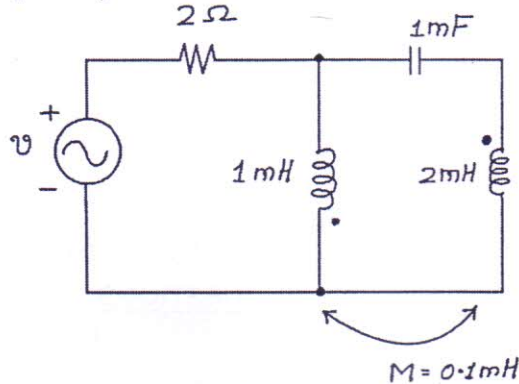


Fig. 3b.

- c) A 3-phase, 3-wire, 240V delta connected RYB system supplies a delta connected load in which $Z_{RY} = 25 \angle 90^\circ \Omega$, $Z_{YB} = 15 \angle 30^\circ \Omega$ and $Z_{BR} = 20 \angle 0^\circ \Omega$. Find the line currents.

Unit – II

4. a) Obtain Laplace Transform of the signal shown in Fig. 4a.

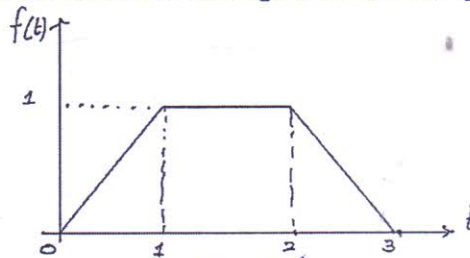


Fig. 4a.

- b) In the circuit in Fig. 4b. the switch is changed from position 1 to the position 2 at $t = 0$, steady condition having reached before switching. Find the values of i , $\frac{di}{dt}$ and $\frac{d^2i}{dt^2}$ at $t = 0^+$

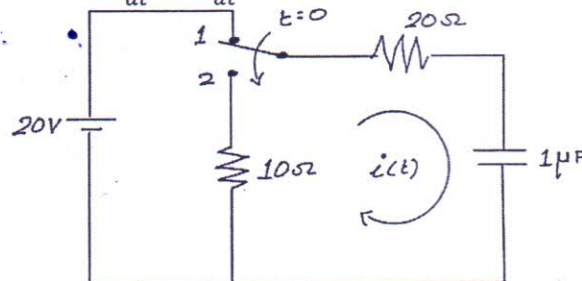


Fig. 4b.

- c) The network in Fig. 4c. has acquired steady state with the switch closed for $t < 0$. At $t = 0$, the switch is opened. Obtain $i(t)$ for $t > 0$.

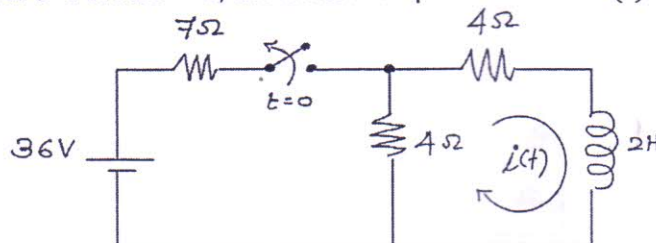


Fig. 4c.

5. a) Obtain the Laplace Transform of the signal shown in Fig. 5a.

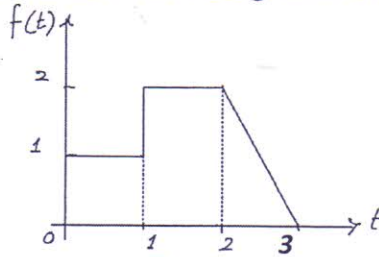


Fig. 5a.

- b) In the circuit shown in Fig. 5b. the switch K is opened at time $t = 0$ after reaching the steady state. Find v_k and $\frac{dv_k}{dt}$ at time $t = 0^+$.

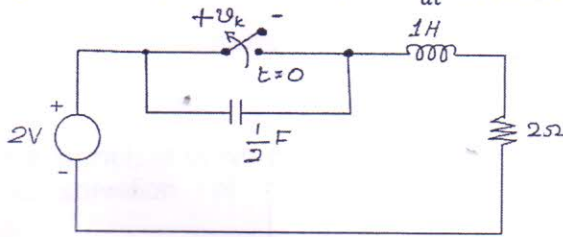


Fig. 5b.

- c) Find the h-parameters of the network shown in Fig. 5c.

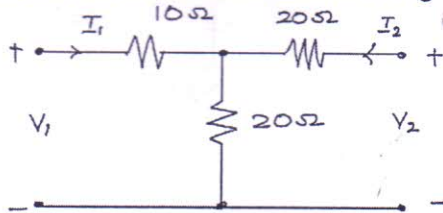


Fig. 5c.

6. a) Find the y-parameters for the network shown in Fig. 6a.

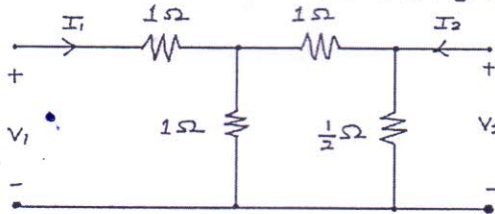


Fig. 6a.

- b) Using s-domain analysis, obtain the expression for current in a series RLC circuit for $\alpha < \omega_0$.

Unit – III

7. a) Find the Thevenin equivalent circuit between the terminals A and B for the circuit in Fig. 7a. Find the value of R so that the maximum power is delivered to the load. Also compute the maximum power delivered to the load.

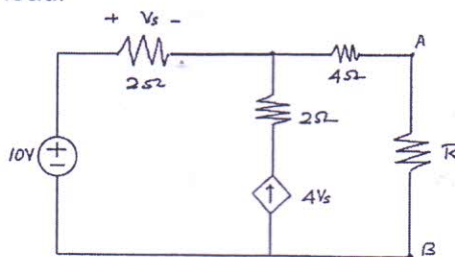


Fig. 7a.

4 L3 3 1,2

6 L3 3 1,2

6 L3 4 1,2

8 L3 4 1,2

8 L3 4 1,2

8 L3 5 1,2

- b) Find the voltmeter reading in the circuit shown in Fig. 7b. Validate the reciprocity theorem by interchanging the positions of voltmeter and the current source.

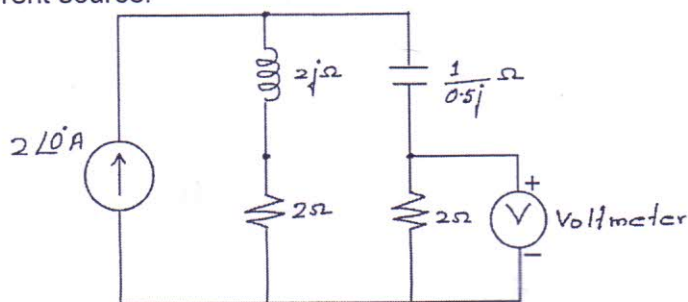


Fig. 7b.

8 L3 5 1,2

8. a) For the network in the Fig. 8a. using superposition theorem find current i_0 .

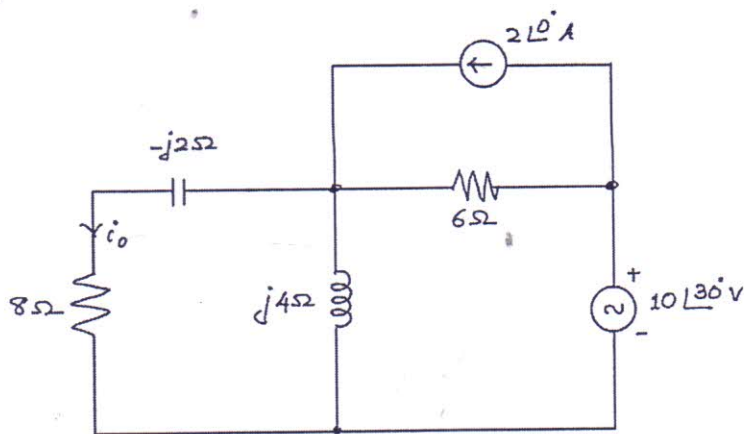


Fig. 8a.

8 L2,L3 5 1,2
8 L3 5 1,2

- b) State and Prove Millman's theorem.

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
